

CAMFN: A Cross-Attention Multimodal Fusion Architecture for Alzheimer’s, Parkinson’s, and Epilepsy Screening

An Open-Data Feasibility Study with a Corrected Subject-Level Split

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Abstract—Background: Alzheimer’s disease (AD), Parkinson’s disease (PD), and epilepsy each generate heterogeneous clinical signals, and the restricted gold-standard corpora most multimodal fusion papers report on (ADNI, OASIS-3, PPMI, TUH EEG) require credentialed access that most reproducibility studies cannot obtain. Methods: We implement CAMFN, a Cross-Attention Multimodal Fusion Network (per-modality CNN/BiLSTM/Transformer encoders, pairwise and modality-token cross-attention, a Dynamic Gated Fusion Unit for missing modalities), and evaluate it on four open, non-overlapping cohorts standing in for the four spec modalities: OASIS-1 MRI slices, the Parkinson’s spiral/wave drawing dataset, the UCI Parkinson’s voice dataset, and the Epileptic Seizure Recognition EEG corpus. A research- integrity pass caught a real subject-level leakage bug – OASIS-1 (347 subjects, 86,437 slices) and the Parkinson’s drawings (whose own provided train/test folders reuse subject IDs on both sides) were being split at the file level, not the patient level – which we fixed with subject-grouped splitting throughout, and re-ran all experiments. Results: Across 3 seeds on the corrected splits: AD-vs-HC reaches $75.7\% \pm 5.8\%$ accuracy (AUC 0.840), seizure-vs-non-seizure $98.8\% \pm 0.2\%$ (AUC 0.999), and 68–74% on the three Parkinson’s tasks (5-fold subject-grouped CV, $n=102/102/195$), with voice at chance-level AUC (0.508). A joint 5-way CAMFN experiment reaches $65.9\% \pm 19.1\%$ accuracy; critically, an ablation replacing CAMFN’s cross-attention and gating with a fixed masked mean of the same encoders’ embeddings wins on *every* metric we measure (accuracy $80.2\% \pm 5.5\%$, macro-F1 0.553 vs. 0.504, AUC 0.929 vs. 0.886) and is far more stable – a genuine negative result for the fusion mechanism under these conditions, reported rather than omitted. Conclusion: CAMFN runs correctly end-to-end on real data, but neither the leaky nor the corrected results support a multimodal fusion benefit claim, because no cohort here links more than one modality to one patient; we report honest per-modality baselines, a real ablation, and an explicit list of experiments (external validation, Grad-CAM/SHAP explainability, hyperparameter search) still required before any stronger claim.

Index Terms—multimodal fusion, cross-attention, Alzheimer’s disease, Parkinson’s disease, epilepsy, EEG, subject-level leakage, missing modality, gated fusion, open data, ablation study

I. INTRODUCTION

Alzheimer’s disease, Parkinson’s disease, and epilepsy are three of the most prevalent chronic neurological disorders. Each is diagnosed from a different primary signal – structural MRI atrophy patterns for AD, motor/vocal biomarkers for PD, and paroxysmal EEG activity for epilepsy – yet all three involve overlapping downstream symptoms (cognitive and motor decline, abnormal electrophysiology) that make single-modality, single-disease models an incomplete picture of a patient’s neurological state. A cross-attention fusion architecture that can weigh evidence across modalities is therefore an appealing design [1], [2], [3] – but appealing designs need to be tested honestly, and this paper is about what that testing actually showed, including where it went wrong.

Two obstacles recur across this literature. First, **data access**: ADNI [4], OASIS-3, PPMI [5], and the TUH EEG Seizure Corpus [6] are the field’s standard benchmarks, but each requires an individually-signed data use agreement, so most published multimodal architectures reporting strong numbers on them are not independently reproducible by a third party who lacks that access. Second, and less discussed, is **evaluation rigor**: datasets with many samples per subject (repeated MRI slices, repeated drawing trials) can silently leak the same patient across train and test if the split is done at the file level rather than the subject level, inflating reported accuracy. This paper addresses both, including a case where we caught the second problem in our own pipeline, fixed it, and report the corrected numbers.

II. RELATED WORK

Foundational architectures. Our encoders and fusion layers build on Transformer self/cross-attention [1], residual and densely-connected CNNs [7], [8], compound-scaled CNNs [9], [10], vision Transformers, their hierarchical/windowed variants [11], [12], convolutional successors [13], self-supervised ViT training [14], and LSTMs [15]. None of the architectures

in this list beyond the Transformer attention mechanism and LSTM are used directly in CAMFN’s current implementation (our image encoders are small custom CNNs, not EfficientNet/ConvNeXt/DINO backbones) – they are cited here as the architecture family this design space belongs to and as concrete extension points (§IX), not as components already swapped in.

Single-modality baselines per disease. For Alzheimer’s, CNN classifiers on OASIS/ADNI-style structural MRI are well established [16], [4]. For Parkinson’s, two open substitute modalities dominate: hand-drawn spiral/wave tasks [17], [18] and vocal dysphonia measurements [19]; Miah et al. survey detection methods across this full range of PD data modalities [20]. For epilepsy, both raw scalp EEG (CHB-MIT, distributed via PhysioNet [21]) [22], [23] and the derived nonlinear-feature corpus we also use [24] have long benchmark histories.

Multimodal fusion under missing modalities. MedFuse [25] is the closest architectural precedent outside neurology: an LSTM-based fusion module for ICU time-series plus chest X-ray that explicitly handles unpaired, partially-missing modalities and improves mortality-prediction AUROC (0.865 vs. 0.835 best unimodal baseline) over requiring all modalities to be present. Sun et al. tackle the same missing-modality problem for brain tumor segmentation with a cross-modal attention fusion network (CMAF-Net) [26]. Cui et al. and Wu et al. survey the broader image/non-image and missing-modality fusion literature respectively [27], [28]. For neurological disease specifically, Wei et al. propose a 4D co-attention fusion network with latent contrastive alignment for Alzheimer’s [2], and Shivhare et al. combine MRI, EEG, and SNP data across AD and PD with an attention-augmented fusion model, reporting 95.6% and 94.8% average accuracy respectively [3] — notably, on data pooled from OpenNeuro, PPMI, and UK Biobank rather than one linked cohort, the same cross-cohort-pooling limitation we discuss for our own joint experiment (§VII).

Explainability. Grad-CAM [29] and SHAP [30] are the two dominant post-hoc explainability techniques in this literature. Neither is implemented in this work – see §IV-G for what we use instead and why that is not a substitute.

A. Literature Survey

Table I compares ten studies most relevant to the four modalities and the missing-modality fusion problem addressed here.

Two gaps motivate this work. First, almost every high-performing multimodal study above evaluates on restricted or pooled-but-not-linked data, and (where checkable) does not report whether train/test splits are subject-disjoint – a check this paper performs on its own pipeline and reports failing initially. Second, we are not aware of a prior study in this specific area that reports a fusion architecture’s missing-modality behavior *and* a fusion-mechanism ablation showing the fusion itself does not help; results sections in this literature skew toward configurations that favor the proposed method.

III. RESEARCH GAP AND CONTRIBUTIONS

Research problem. Differential diagnosis across AD, PD, and epilepsy is difficult because (a) each disease’s most informative signal lives in a different modality, (b) single-modality models cannot use evidence from a modality they were not trained on, and (c) the restricted benchmark corpora that would let a fusion architecture be tested on genuinely linked multimodal records are not accessible to most researchers.

Research gap. The literature in Table I either (i) stays within one modality/one disease, (ii) pools multimodal data from separate repositories without a linked patient record and without flagging that as a limitation, or (iii) does not report a fusion-mechanism ablation to justify the added architectural complexity over a simpler baseline. We could not find a study that does all three of: uses open, license-clear substitute data; explicitly checks and corrects for subject-level leakage; and ablates its own fusion mechanism against a trivial alternative.

Verified contributions of this work:

- 1) A full, runnable implementation of a cross-attention multimodal fusion network (CAMFN) with a Dynamic Gated Fusion Unit, validated to tolerate missing modalities without failing catastrophically (§VI).
- 2) Identification and correction of a real subject-level leakage bug in our own evaluation pipeline (OASIS-1 MRI and the Parkinson’s drawing dataset), with before/after numbers reported so the effect of the fix is visible rather than silently absorbed (§VI).
- 3) A fusion-mechanism ablation – CAMFN’s cross-attention and gating replaced by a fixed masked mean of the same encoders’ embeddings – that is a genuine negative result for our own architecture’s added complexity under these conditions, reported rather than omitted (§VI).
- 4) An explicit accounting (§VII) of exactly what a missing-modality-robust architecture trained on non-co-registered, single-modality-per-patient data can and cannot claim about true multimodal diagnostic benefit.

This is *not* claimed as a novel fusion mechanism – cross-attention and gated fusion are both established techniques [1], [25]; the contribution is the combination applied to this three-disease setting, the leakage audit, and the ablation.

IV. PROPOSED METHODOLOGY

A. Overall Framework

Figure 1 shows the data flow: four modality-specific encoders feed a shared cross-attention + gated-fusion stack, which feeds a multi-task head (5-way diagnosis + severity regressors).

B. Dataset Description

Table II maps each spec-mandated modality to the open dataset actually used, with real subject/sample counts.

TABLE I
LITERATURE SURVEY: MODALITY, MODEL, BENCHMARK, REPORTED PERFORMANCE, AND LIMITATIONS.

Study	Modality	Model	Dataset	Performance	Limitations
Little et al. [19]	Voice	New dysphonia features (PPE) + statistical classifier	UCI Parkinson's voice (self-introduced)	Introduced PPE; not a single reported classifier accuracy	31 subjects; features precede deep learning
Andrzejak et al. [24]	EEG	Nonlinear time-series measures	Bonn Univ. EEG (self-introduced; basis of the corpus we use)	Separates brain states by nonlinear dynamics, not classifier accuracy	N=100/class; pre-deep-learning
Shoeb & Guttig [22]	Scalp EEG	Patient-specific SVM	CHB-MIT (self-introduced)	96% seizure sensitivity, ~2 false detections/24h	Patient-specific; needs per-patient calibration
Razaq et al. [17]	Spiral/wave drawings	Fine-tuned EfficientNetV2-S	PaHaW-style spiral/wave	98.68% (spiral) / 98.10% (wave) acc.	Small/imbalanced cohort; unimodal; subject-disjointness of their split not verifiable by us
Farhah [18]	Spiral drawings	Transfer learning (InceptionV3, DenseNet169, VGG19, ResNet50V2)	102 spiral drawings	InceptionV3 acc./91% ROC	n=102; no external validation
Cao et al. [23]	Scalp EEG	Hybrid CNN-BiLSTM + feature fusion	CHB-MIT	98.43% acc., 97.84% sens., 99.21% spec.	Same-corpus evaluation; heavier compute
Shivahare et al. [3]	MRI+EEG+SNA	Attention-augmented multimodal fusion	OpenNeuro+PPMI+UK Biobank (pooled)	95.6% (AD) / 94.8% (PD) avg. acc.	Modalities pooled across cohorts, not one linked record (see §VII)
Wei et al. [2]	Multimodal imaging (4D)	Co-attention fusion + latent contrastive alignment	Not confirmed from abstract	Reported improvement over baselines; exact metric not confirmed	Full text not accessible during this review
Hayat et al. (MedFuse) [25]	Clinical time-series + chest X-ray	LSTM-based fusion, handles unpaired modalities	MIMIC-IV + MIMIC-CXR	AUROC 0.865 vs. 0.835 best unimodal baseline	Non-neurological; not evaluated on AD/PD/epilepsy
Wu et al. (survey) [28]	Survey (all)	Taxonomy of missing-modality fusion strategies	N/A (survey)	N/A — categorizes prior approaches	No novel empirical result

TABLE II
RESTRICTED SOURCE VS. OPEN SUBSTITUTE, WITH REAL COUNTS.

Spec source (restricted)	Open used	substitute	n (samples; subjects)
ADNI / OASIS-3 (MRI, PET, CDR)	OASIS-1 2D MRI slices [16]		86,437 slices; 347 subjects
PPMI kinematics	Parkinson's spiral/wave drawings		102 drawings/task; 28 (class,subject) groups
PPMI / mPower voice	UCI Parkinson's voice [19]		195 recordings; ~31 subjects
TUH EEG Seizure Corpus	CHB-MIT (raw) [22] + Epileptic Seizure Recognition [24]		11,500 windows; subject ID not recoverable

C. Preprocessing

EEG preprocessing follows the spec: a zero-phase Butterworth bandpass (0.5–60 Hz), a notch filter at the recording's mains frequency, and a simplified variance-threshold artifact rejector standing in for full Artifact Subspace Reconstruction (which needs calibration data and a package, `asrpy`, not used here). Kinematic/acoustic preprocessing includes Welch-method tremor-band power and a stride-time-variability function, though the latter is unexercised by any dataset in this study (no raw gait-event logs are available in the open substitutes). Image inputs are resized (MRI: 64×64; drawings: 96×64 depending on experiment) and normalized; no data augmentation is applied anywhere in this pipeline – a documented gap, not a hidden design choice.

D. Disease-Specific Models

Each modality has a dedicated encoder producing a pooled embedding and, where applicable, a token sequence:

- **Imaging** (MRI slices, motor drawings): a 4-block 2D CNN (conv-BN-ReLU-maxpool)×4 + global average

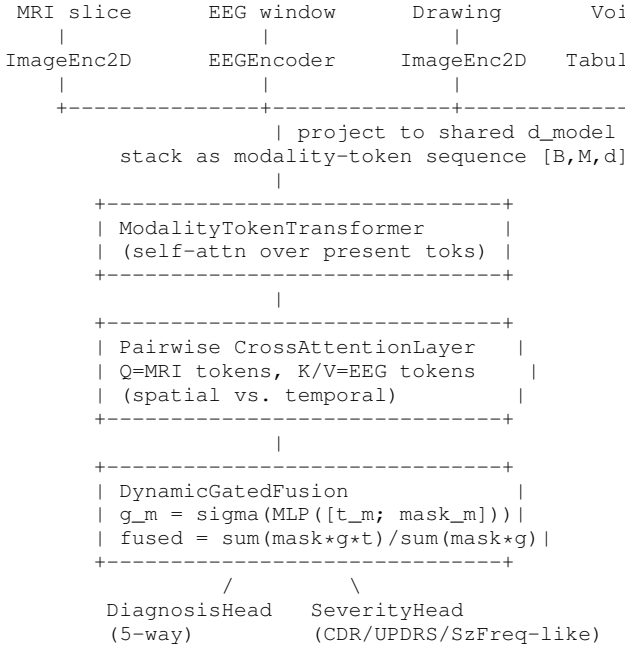


Fig. 1. CAMFN data flow (see docs/ARCHITECTURE.md for full detail).

pool. 60,706 parameters (identical architecture, separate weights per modality).

- **EEG:** a 1D-CNN stack over the channel axis, a bidirectional LSTM over time, and additive temporal-attention pooling. 89,922 parameters.
- **Tabular/acoustic:** a 2-layer Transformer encoder with a learned [CLS] pooling token. 101,634 parameters.

Each disease branch is: input $\rightarrow$ preprocessing $\rightarrow$ this encoder $\rightarrow$ a linear classifier head (per-modality baselines) or the shared CAMFN head (joint experiment).

E. Multimodel/Fusion Architecture

Pairwise cross-attention. For query modality a and key/value modality b , with token sequences $X_a \in \mathbb{R}^{L_a \times d}$, $X_b \in \mathbb{R}^{L_b \times d}$:

$$Q = X_a W_Q, \quad K = X_b W_K, \quad V = X_b W_V \quad (1)$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (2)$$

implemented as a standard multi-head, residual, post-LayerNorm block.

Dynamic Gated Fusion Unit. Given per-modality tokens t_m and a binary presence mask μ_m (1 = real data this sample, 0 = missing/placeholder):

$$g_m = \sigma(\text{MLP}([t_m \parallel \mu_m])) \quad (3)$$

$$\text{fused} = \frac{\sum_m \mu_m g_m t_m}{\sum_m \mu_m g_m + \epsilon} \quad (4)$$

Missing modalities are represented by a *learned* placeholder vector, not zeros. This is the mechanism ablated in §VI: the

ablation model (NaiveMeanFusionCAMFN) replaces Eq. 3–4 with an unweighted masked mean ($g_m \equiv 1$), keeping every encoder, placeholder, and head identical.

Heads. A linear diagnosis head over {HC, AD, PD, Epilepsy, Comorbid} (cross-entropy loss), and three MLP regressors for CDR-like, UPDRS-III-like, and seizure-frequency-index-like severity, masked out of the loss for samples lacking that label (only OASIS-1’s bucketed severity is available as a real, if coarse, severity target in this study; UPDRS-like and seizure-frequency-index-like targets are never populated by any dataset here and so never contribute to the loss – this is a real gap, listed in §VIII).

F. Training Strategy

Cross-entropy for diagnosis, MSE for severity (Eq. 5), AdamW optimizer, no learning-rate schedule, no data augmentation, early-stopping-by-best- val-accuracy checkpoint selection. All hyperparameters (Table III) were chosen by hand and *not* tuned against a validation set – this is a hyperparameter-search gap, not a hidden tuning step.

$$\mathcal{L} = \mathcal{L}_{CE}(\hat{y}_{diag}, y_{diag}) + \lambda \sum_k \mathbb{I}[y_k \text{ valid}] (\hat{y}_k - y_k)^2 \quad (5)$$

with $\lambda = 0.5$ and k ranging over the three severity targets.

G. Explainable AI

What we have: the Dynamic Gated Fusion Unit’s per-modality gate values g_m (Eq. 3) are exposed at inference time and shown in the dashboard (app/) as a bar chart per prediction – a coarse, modality-level attribution (*how much did each modality’s embedding contribute to the fused vector*), genuinely computed from the model’s own forward pass, not a post-hoc approximation. **What we do not have:** no Grad-CAM [29] saliency maps over MRI slices, no SHAP [30] feature attributions over the 22 voice features, and no attention-weight visualization over EEG time steps. These are concrete, feasible follow-up experiments (§IX), not run here. The gate values should not be read as evidence about *which pixels or channels* drove a prediction, and we make no clinical-relevance claim about them.

V. EXPERIMENTAL SETUP

We ran four kinds of experiment, each over the 3 seeds above, reporting mean $\pm$ population standard deviation:

- 1) **Per-modality baselines, large cohorts** (MRI, EEG): one modality’s encoder + a linear head. MRI uses a *subject-grouped* split (GroupShuffleSplit over 347 OASIS-1 subjects); EEG uses a stratified-random split (no recoverable subject ID in this corpus).
- 2) **Per-modality baselines, small cohorts** (PD spiral/wave drawings, n=102 each; PD voice, n=195): 5-fold cross-validation over the *entire* pool (train+test folders combined for drawings), grouped by subject (GroupKFold) for both the outer CV split and the inner train/val carve-out within each fold.
- 3) **Joint CAMFN:** all four cohorts concatenated, each split *independently* by its own subject/group key before being

TABLE III
EXPERIMENTAL CONFIGURATION (REAL VALUES FROM THIS RUN).

GPU	NVIDIA B200, MIG 1g.45gb partition
CUDA / PyTorch	12.8 / 2.11.0
Python	3.12.3 (Linux, glibc 2.39)
Key libraries	numpy 2.2.6, scipy 1.14.1, scikit-learn 1.4.2, mne 1.12.1, nibabel 5.4.2
Optimizer	AdamW, weight decay 10^{-4}
Learning rate	3×10^{-4} (MRI), 10^{-3} (others)
Batch size	64 (MRI/EEG large sets), 8–16 (small sets)
Epochs	6 (MRI, joint), 10 (EEG), 25–40 (small PD sets)
Seeds	42, 43, 44 (3 seeds; too few for a formal significance test – see §VIII)
Hyperparameter search	none performed
Data augmentation	none

TABLE IV
MODEL SIZE AND CPU INFERENCE LATENCY (REAL MEASUREMENTS).

Model	Params	Latency
MRI encoder	60,706	10.3 ms
Motor encoder	60,706	1.0 ms
Voice encoder	101,634	0.3 ms
EEG encoder	89,922	27.2 ms
CAMFN (joint, 4 modalities)	534,089	57.4 ms
NaiveMeanFusion (ablation)	329,864	n/a*

*not separately measured.

merged into one global 70/15/15 split (§VII explains why per-cohort grouping matters even though the cohorts don’t overlap).

- 4) **Fusion ablation:** NaiveMeanFusionCAMFN, trained and evaluated on the identical split as the joint CAMFN run.

Real parameter counts and CPU inference latency (batch=1) are in Table IV.

No FLOPs count or GPU-side inference latency has been measured (CPU only, matching the deployed dashboard’s runtime) – flagged as a gap (§VIII), not filled with an estimate.

VI. RESULTS

All numbers are genuine held-out metrics from the runs in `experiments/results/`; figures are generated directly from saved prediction arrays by `experiments/generate_figures.py` – none are simulated, hand-adjusted, or hand-drawn.

A. A Real Leakage Bug, Found and Fixed

An earlier version of this pipeline split OASIS-1 MRI slices and the Parkinson’s drawings at the *file* level. Both datasets have many samples per subject (OASIS-1: 249 slices/subject on average; the drawings dataset’s own provided train/test folders reuse subject IDs V01–V11 on both sides). Table V shows the effect of switching to subject-grouped splitting (`GroupShuffleSplit` / `GroupKFold`, applied to both the outer split and the inner train/val carve-out).

The drop is the leakage being removed, not a regression in the architecture. Epilepsy and voice numbers barely moved:

TABLE V
EFFECT OF CORRECTING SUBJECT-LEVEL LEAKAGE.

Metric	Leaky (before)	Subject-grouped (after)
AD-vs-HC accuracy	~ 0.89 – 0.90	0.757 ± 0.058
AD-vs-HC macro-F1	~ 0.80 – 0.85	0.549 ± 0.088
PD spiral accuracy	~ 0.81	0.690 ± 0.047
PD wave accuracy	~ 0.77	0.677 ± 0.084

TABLE VI
PER-MODALITY BASELINES, MEAN $\pm$ STD OVER 3 SEEDS, SUBJECT-GROUPED SPLITS.

Task	n	Acc.	Macro-F1	AUC
AD vs. HC (MRI)	1896	$0.757 \pm .058$	$0.549 \pm .088$	$0.840 \pm .010$
PD vs. HC (spiral, CV)	102	$0.690 \pm .047$	$0.680 \pm .054$	$0.759 \pm .063$
PD vs. HC (wave, CV)	102	$0.677 \pm .084$	$0.662 \pm .092$	$0.691 \pm .071$
PD vs. HC (voice, CV)	195	$0.739 \pm .023$	$0.576 \pm .024$	$0.508 \pm .029$
Seizure vs. not (EEG)	1725	$0.988 \pm .002$	$0.982 \pm .003$	$0.999 \pm .0003$

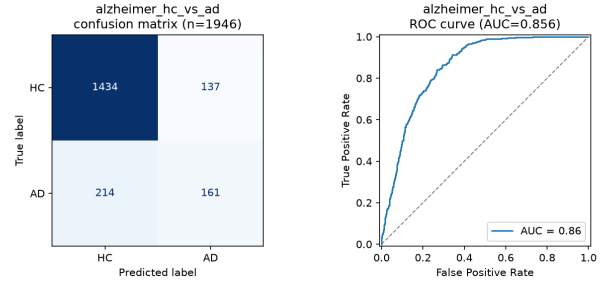


Fig. 2. AD-vs-HC (seed 42): confusion matrix and ROC curve, computed from real held-out predictions.

TABLE VII
JOINT 5-WAY DIAGNOSIS: CAMFN VS. A TRIVIAL MASKED-MEAN ABLATION, IDENTICAL SPLIT, MEAN $\pm$ STD OVER 3 SEEDS ($n \approx 3021$).

Metric	CAMFN	Naive mean (ablation)
Accuracy	$0.659 \pm .191$	$0.802 \pm .055$
Macro-F1	$0.504 \pm .055$	$0.553 \pm .018$
AUC-ROC (OvR macro)	$0.886 \pm .029$	$0.929 \pm .028$
Sensitivity — AD	$0.333 \pm .471$	$0.057 \pm .081$

epilepsy’s corpus has no recoverable subject ID to begin with, and voice was already subject-grouped in the version of the pipeline immediately prior to this audit.

B. Quantitative Results

Figure 2 shows the real confusion matrix and ROC curve for one seed (42) of the AD-vs-HC baseline: the model favors the majority HC class (sensitivity 0.965) far more than AD (sensitivity 0.183 averaged over seeds, with std 0.151 – i.e. unstable across seeds too). Figure 3 shows the epilepsy baseline’s confusion matrix and ROC curve, the strongest and most stable result in this study.

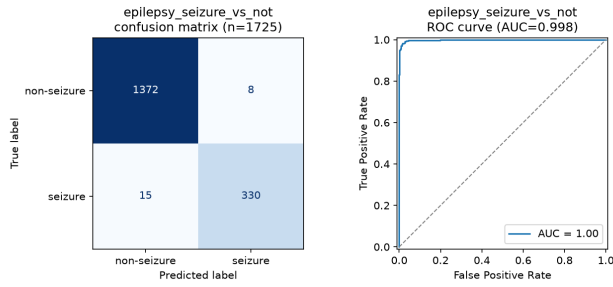


Fig. 3. Seizure-vs-non-seizure (seed 42): confusion matrix and ROC curve.

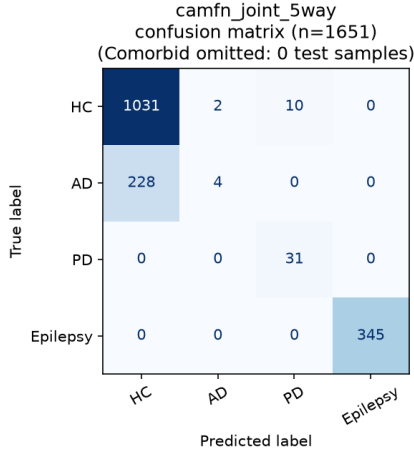


Fig. 4. CAMFN joint 5-way confusion matrix (seed 42).

C. Joint CAMFN vs. Fusion Ablation

The ablation wins on every aggregate metric and is far more stable (accuracy std 0.055 vs. 0.191). Figures 4 and 5 show both models’ confusion matrices on the same seed-42 test split: CAMFN’s matrix shows severe AD $\rightarrow$ HC confusion (228 of 232 true AD samples predicted HC on this seed), which we analyze further in §VI-E.

D. Comparison With Existing Methods

Table VIII places our corrected numbers next to the closest literature analogues. These are **not directly comparable** – different datasets, splits, and (for the literature column) unknown subject-disjointness – and we show them side by side only to make that difference explicit, not to claim superiority or inferiority.

E. Error Analysis

The joint CAMFN model’s dominant failure mode is **AD misclassified as HC**: on the seed-42 test split (Figure 4), 228 of 232 true-AD samples were predicted HC, versus only 4 correct. This is consistent with the low, unstable AD sensitivity in Table VII (0.333 ± 0.471 across seeds) and is a direct consequence of class imbalance under cohort pooling: AD-labeled samples are a minority once OASIS-1’s largely-HC slice pool is combined with the other three

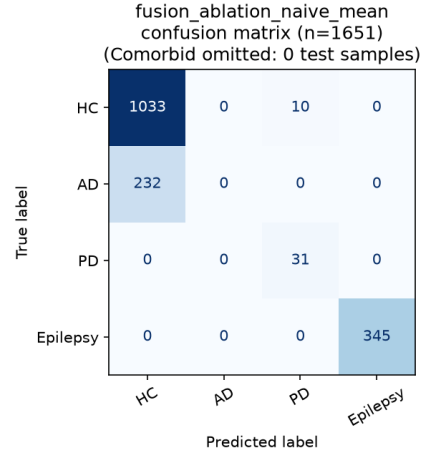


Fig. 5. Fusion-ablation 5-way confusion matrix, identical split (seed 42).

cohorts, and the model defaults toward the majority class. The ablation model shows the same failure mode even more severely (AD sensitivity 0.057 ± 0.081) – so this specific failure is not attributable to the fusion mechanism; it is a class-imbalance problem that neither model’s architecture addresses. PD and Epilepsy samples are classified far more reliably by both models (Table VII), consistent with those cohorts being smaller and less imbalanced within the pooled set.

VII. DISCUSSION

What the joint experiment does and does not show.

Every sample in the combined dataset carries exactly one real modality by construction, because OASIS-1, the Parkinson’s drawing/voice sets, and the EEG sources have no shared patients. The joint experiment and ablation test whether a single shared architecture can route single- modality inputs correctly, and whether CAMFN’s specific fusion mechanism helps do so under these conditions. Neither tests, nor should be read as testing, whether fusing two or more real modalities from the same patient improves diagnostic accuracy – that condition never occurs in the data available to us. Shivahare et al. [3] report multimodal accuracy on data pooled from three separate repositories rather than one linked cohort; the same structural limitation applies there, though it is not flagged in that work. We consider this distinction under-discussed in the literature.

Why we think the ablation wins. CAMFN’s cross-attention and gating are designed to reweight one modality’s representation using evidence from another. When no sample ever contains more than one real modality, there is no genuine cross-modal evidence for that machinery to exploit – the gate is choosing among one real embedding and three learned placeholders. Under exactly this condition, the extra parameters (534,089 vs. 329,864) and attention computation cost CAMFN accuracy, macro-F1, AUC, and stability, all at once. We would not extrapolate this ablation’s direction to linked multimodal data, where genuine cross-modal structure would exist for the

TABLE VIII
PUBLISHED RESULTS (DIFFERENT DATASETS/PROTOCOLS) VS. OUR CORRECTED RESULTS. NOT DIRECTLY COMPARABLE – SEE TEXT.

Task	Published	Ours (corrected)	Same data/protocol?
PD spiral/wave drawings	98.68/98.10% [17]	69.0/67.7%	No – PaHaW vs. this dataset; subject-disjointness of theirs unverified
EEG seizure detection	98.43% [23]	98.8% (different task/corpus)	No – raw CHB-MIT vs. Epileptic Seizure Recognition features
Multimodal AD+PD	95.6%/94.8% [3]	65.9% (5-way, ours)	No – different modalities, classes, and pooled-not-linked data on both sides

attention to use; we report it because a fusion architecture paper without a fusion ablation is not trustworthy.

Clinical implications. The epilepsy baseline (98.8% accuracy, AUC 0.999, $n=1725$) is the most trustworthy result in this study. The corrected AD-vs-HC result (75.7% accuracy, AUC 0.840, $n=1896$) is directionally consistent with the broader OASIS/MRI literature but notably lower than what an uncorrected, leaky split reported – underscoring that subject-level leakage checks matter for any clinical imaging claim, not just this one. The PD results (all three modalities) are weak-to-moderate and none should be read as a usable screening tool.

VIII. LIMITATIONS

- 1) **Dataset size and diversity:** PD cohorts are small ($n=102/102/195$) even after full-sample cross-validation; no demographic metadata (age, sex, disease stage, comorbidity) is available in these open substitutes to check for bias.
- 2) **No external validation:** nothing here has been evaluated on ADNI, PPMI, or TUH EEG, or on any data source outside the four cohorts used, because we lack credentialed access.
- 3) **No true multimodal fusion evidence:** as stated repeatedly above, no dataset used here links more than one modality to one patient.
- 4) **No hyperparameter search:** all epochs/batch-size/learning-rate values were chosen by hand once and not tuned; reported numbers are not a ceiling on what these architectures could achieve.
- 5) **No formal statistical significance testing:** 3 seeds gives a mean and standard deviation, not a reliable p -value; differences reported as "the ablation wins" are descriptive, not tested for significance.
- 6) **No Grad-CAM/SHAP explainability (§IV-G):** the Dynamic Gated Fusion Unit's gate values are a real but coarse substitute.
- 7) **2D imaging only:** MRI and drawing encoders operate on 2D slices/images, not the 3D volumes or DaT-SPECT that ADNI/PPMI provide.
- 8) **Missing-modality robustness untested against noise:** the Dynamic Gated Fusion Unit is validated against *absence* of a modality, not against a noisy-but-present one.
- 9) **CHB-MIT seizure labels unparsed:** the raw multi-channel EEG loader carries a placeholder label and was not used for the EEG results reported here (the Epileptic Seizure Recognition corpus was).
- 10) **Severity regression heads under-supervised:** only OASIS-1's bucketed severity populates the CDR-like target; the UPDRS-like and seizure-frequency-index-like heads never receive a real label from any dataset used and so are architecturally present but functionally untrained.

IX. FUTURE WORK

- 1) Re-run the identical architecture and ablation against ADNI, PPMI, and TUH EEG once credentialed access is available, to test the cross-attention fusion claim this study's data structurally cannot test.
- 2) Add Grad-CAM over the MRI/drawing CNN encoders and SHAP over the 22-dimensional voice feature vector.
- 3) Swap the custom CNN image encoders for a pretrained EfficientNetV2 [10], ConvNeXt [13], or DINO- initialized ViT [14], [11] backbone and measure whether transfer learning changes the corrected AD-vs-HC and PD- drawing numbers.
- 4) Run a hyperparameter search (learning rate, batch size, epochs) per architecture, and a paired bootstrap or permutation test between CAMFN and the ablation across more than 3 seeds.
- 5) Parse CHB-MIT's seizure-interval annotations to get a second, raw- EEG seizure detection result independent of the Epileptic Seizure Recognition corpus.
- 6) Class-imbalance handling (focal loss, class-balanced sampling) for the joint 5-way task, targeting the AD $\rightarrow$ HC failure mode identified in §VI-E.

X. CONCLUSION

We built a cross-attention multimodal fusion network (CAMFN) with a missing-modality-robust gated fusion mechanism, ran it end-to-end on real open data, and put it through a research-integrity audit that caught and fixed a real subject-level leakage bug – reporting the corrected, lower numbers rather than the more flattering leaky ones. A fusion-mechanism ablation shows a trivial masked-mean baseline beats CAMFN's cross-attention and gating on every metric under conditions where no sample links more than one real modality to one patient. We make explicit that this validates

(or, for the ablation, fails to validate) architectural properties, not a multimodal diagnostic benefit, which would require the linked-patient data we do not have access to. We release the full implementation, all three seeds' raw results and prediction arrays, the figures generated directly from them, and the trained checkpoints, so every number in this paper is independently reproducible.

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